

Luxury Cosmetics Pop-up Events Analysis

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Harshada S. (Harsha)

Data Analytics Certification Course | Brainstation.

Today's Agenda



- Introduction
- Business Objective & why
- Hypothesis
- Data preparation & Refinement
- Some Stats and Outliers
- Hypothesis Analysis & Insights
 - *Airport Vs Non-airport Conversion Analysis*
 - *Dashboard - for Reference*
- Key Findings
- Conclusion
- Recommendation
- Thank you

Introduction

LUXURY COSMETICS POP-UP EVENTS & ITS ANALYSIS (FEB 2024 - AUG 2025)

Luxury Cosmetic Brands increasingly rely on **pop-up retail events** across different location types

Location Type	Brands	Regions & Cities
Airport Duty Free	24 Different Luxury Brands	5 Different Regions
Luxury malls		26 Unique Cities
High Streets		
Art & Design Districts		
Department store Atriums		

Business Problem & Why

To analyze **2024–2025 pop-up events' performance** using sales, footfall, and conversion metrics to support **future location decisions**

- To better invest in marketing and pop-up events campaigns
- To increase conversion
- Expensive to run, especially in luxury product market.
- Most argue— Airport Dutyfree has more footfall/ no. of visitors - might give more conversion

❖ Objective:

Evaluate whether **Airport Duty-Free pop-ups** perform better than **Non-Airport pop-ups**.

❖ Hypothesis

- **H₀ (Null):** No performance difference
- **H₁ (Alternate):** Non-airport pop-ups have higher conversion efficiency than Airport duty-free location type.

❖ Key KPI Used:

Conversion Rate = Units Sold ÷ Average Daily Footfall

DATA PREPARATION & REFINEMENT

Dataset :

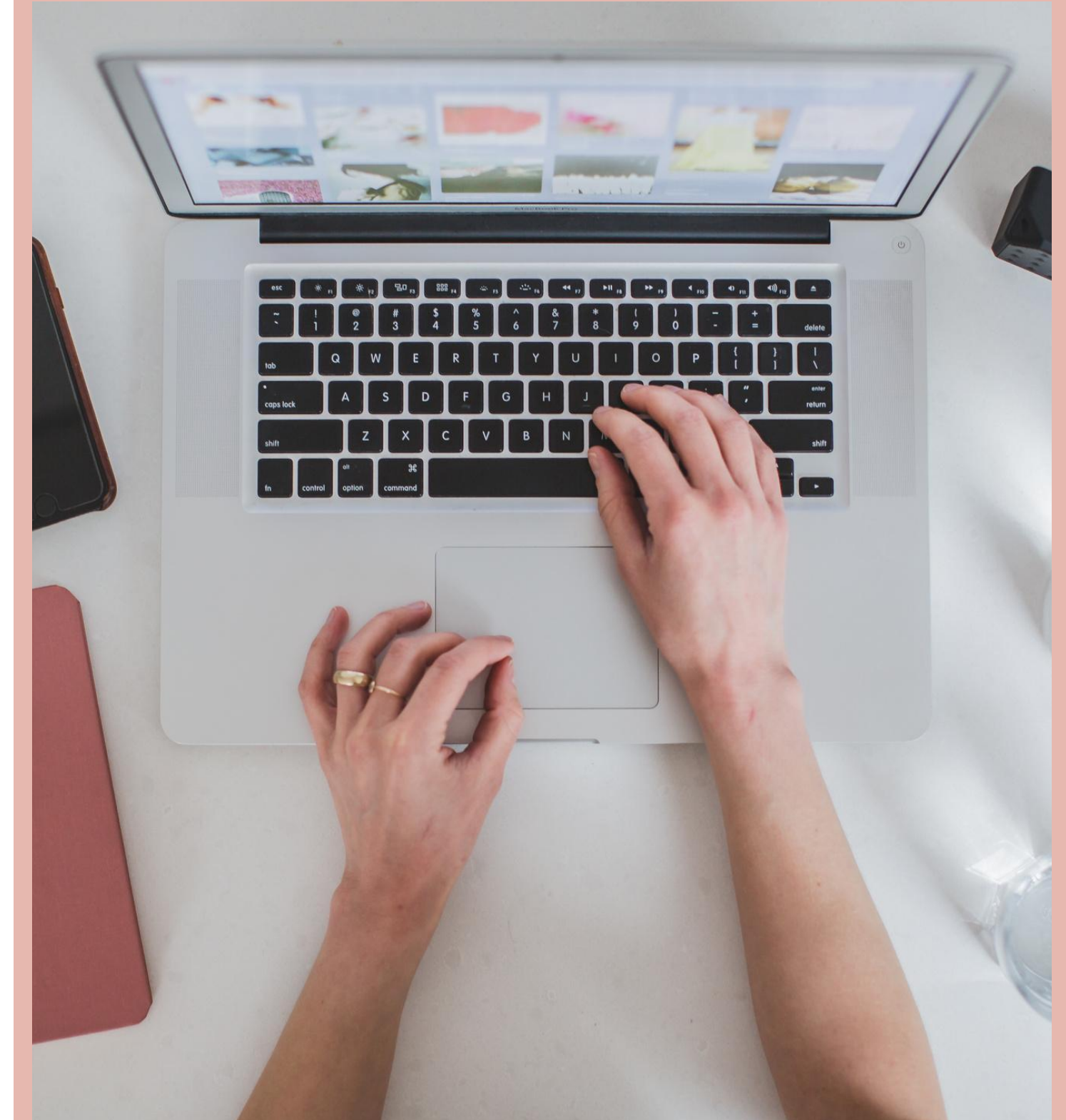
Luxury cosmetics pop-up events (2024–2025) From Kaggle.

Data Refinement Process & Analysis

(Excel, PostgreSQL, Tableau Public, Dbeaver, Chatgpt, Canva)

- **Manages records with:**
 - Zero footfall
 - Missing sales values
 - Date format from Strings to Date
 - Blank cities to Others
- Standardized location types
- **Created new fields:**
 - Total Revenue
 - Conversion Rate
 - Weighted Conversion Rate
 - Location Group (Airport vs Non-Airport)
 - Product Category (Fragrance / Make-up)

✅ Result: Clean, analysis-ready dataset



Some Statistics & Outlier

Calculations-

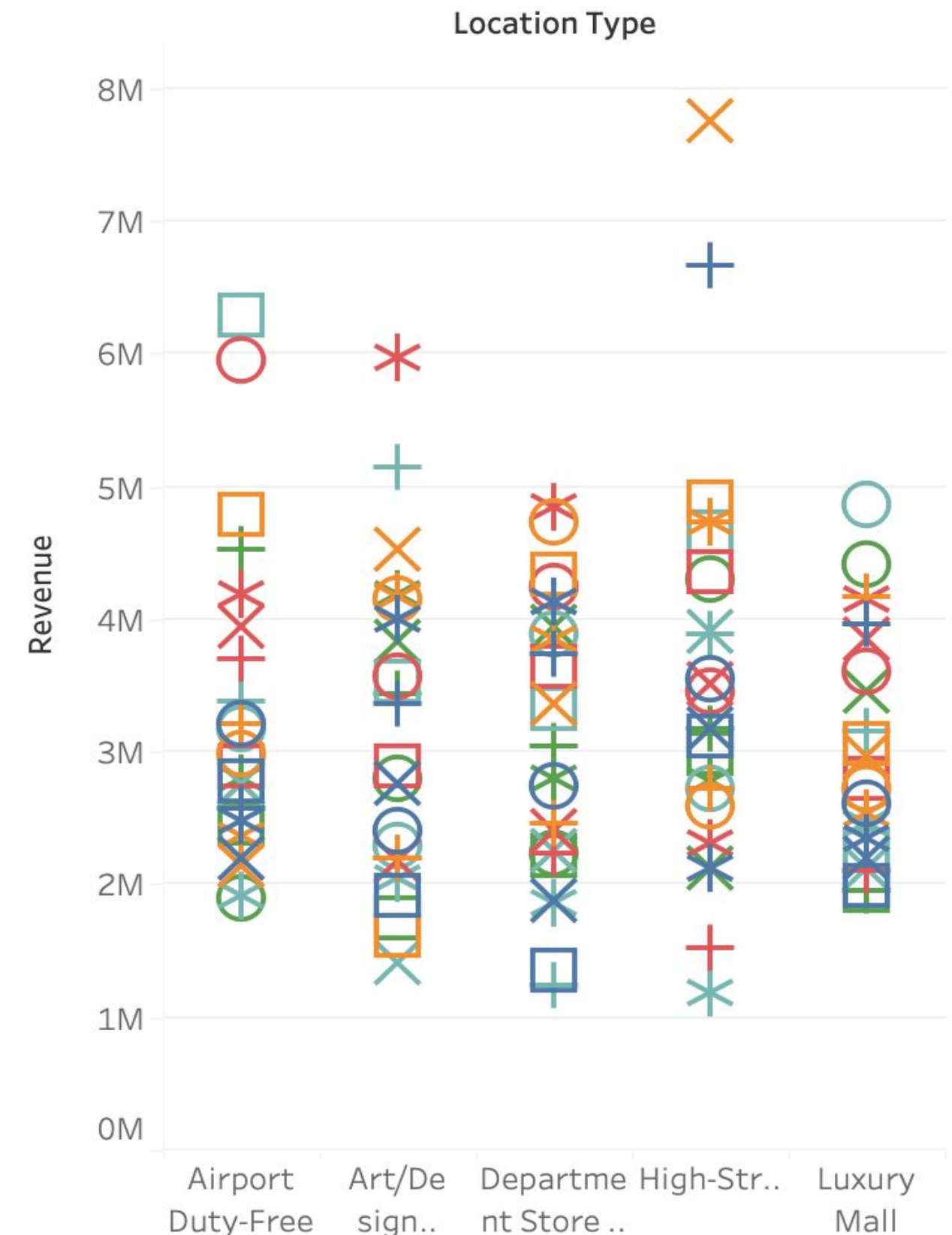
- Total Revenue, Avg Revenue, sales conversion Rate, Weighted & median Conversion Rate, its mean and more

Outlier Identification -

- Used Average, Mean and Standard Deviation
- **Flagged:**
 - Extremely high footfall
 - Very high conversion rates

Assumption

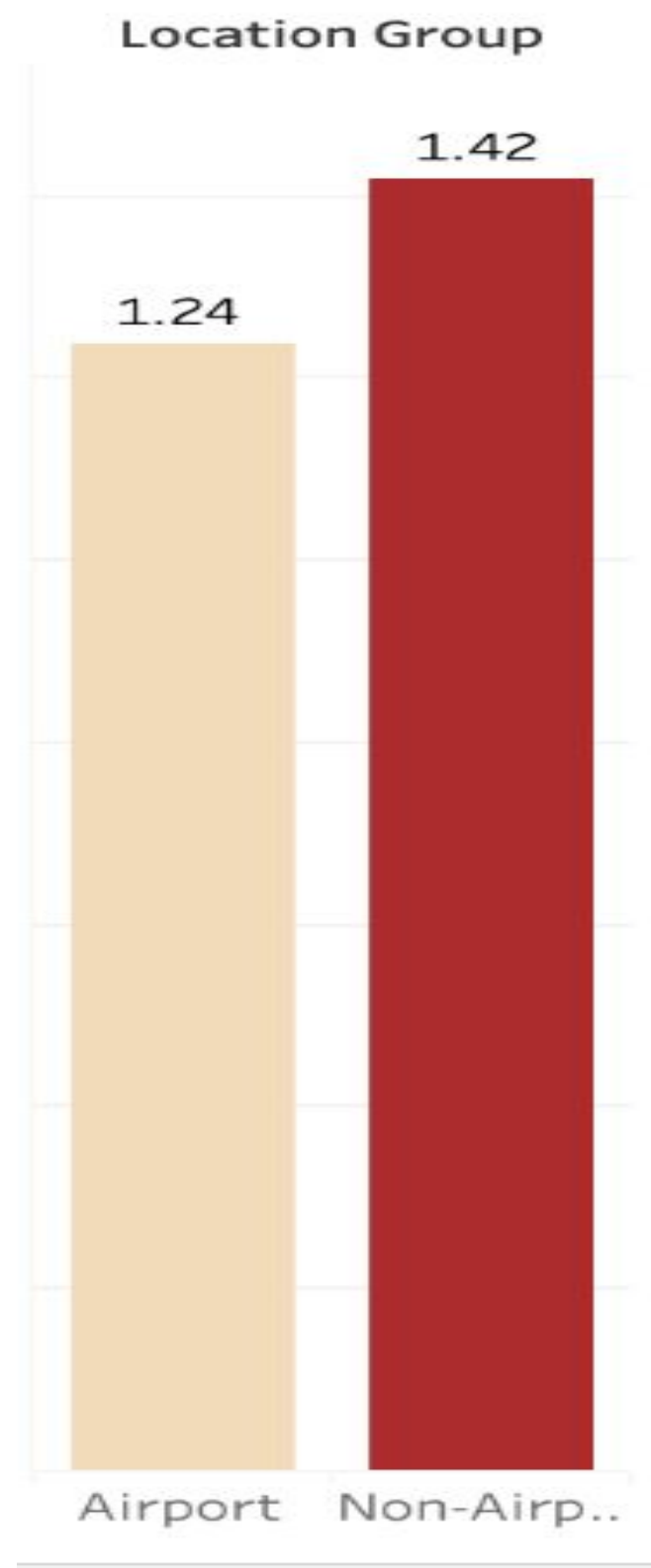
- Outliers were not removed
- **Reason:**
 - Luxury pop-ups naturally show high variance
 - Removing them would hide real business behavior



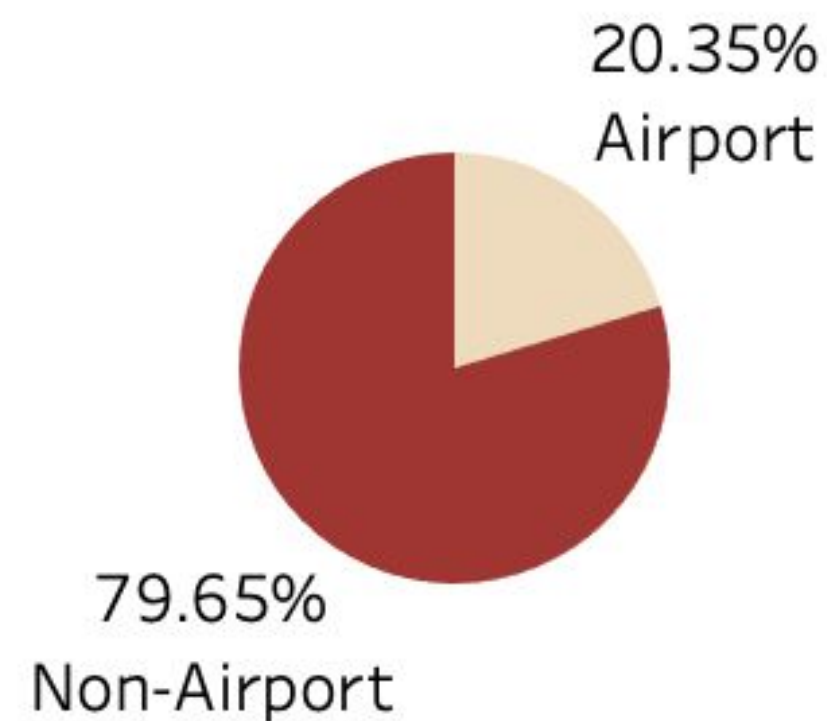
Hypothesis Analysis

Airport
vs
Non-Airport
Conversion Analysis
Feb 2024 - Aug 2025

**WEIGHTED CONVERSION
RATE**



EVENT DISTRIBUTION REVENUE-WISE



Airport

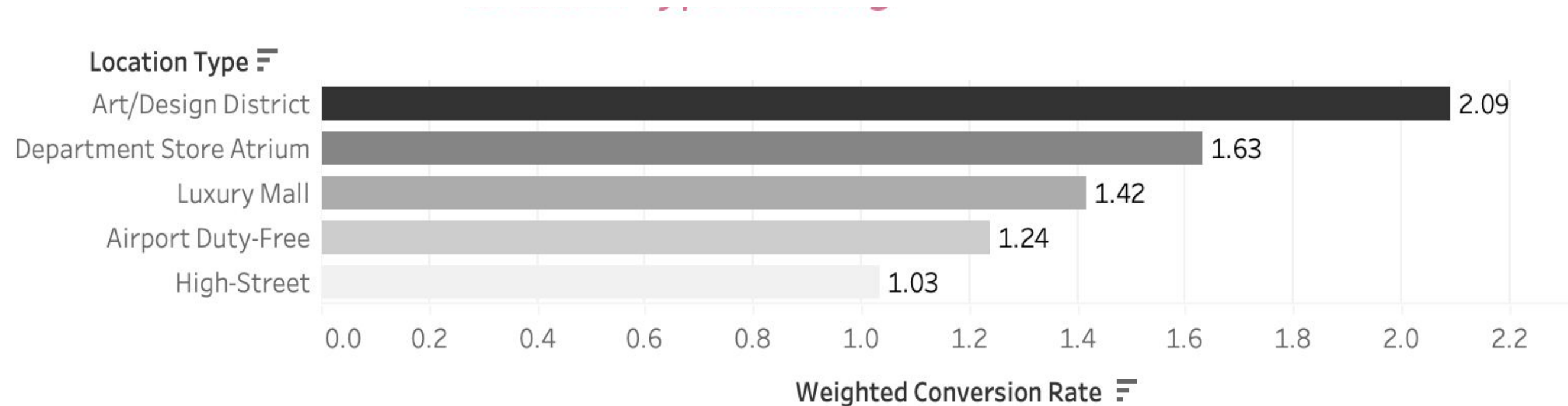
vs

Non-Airport

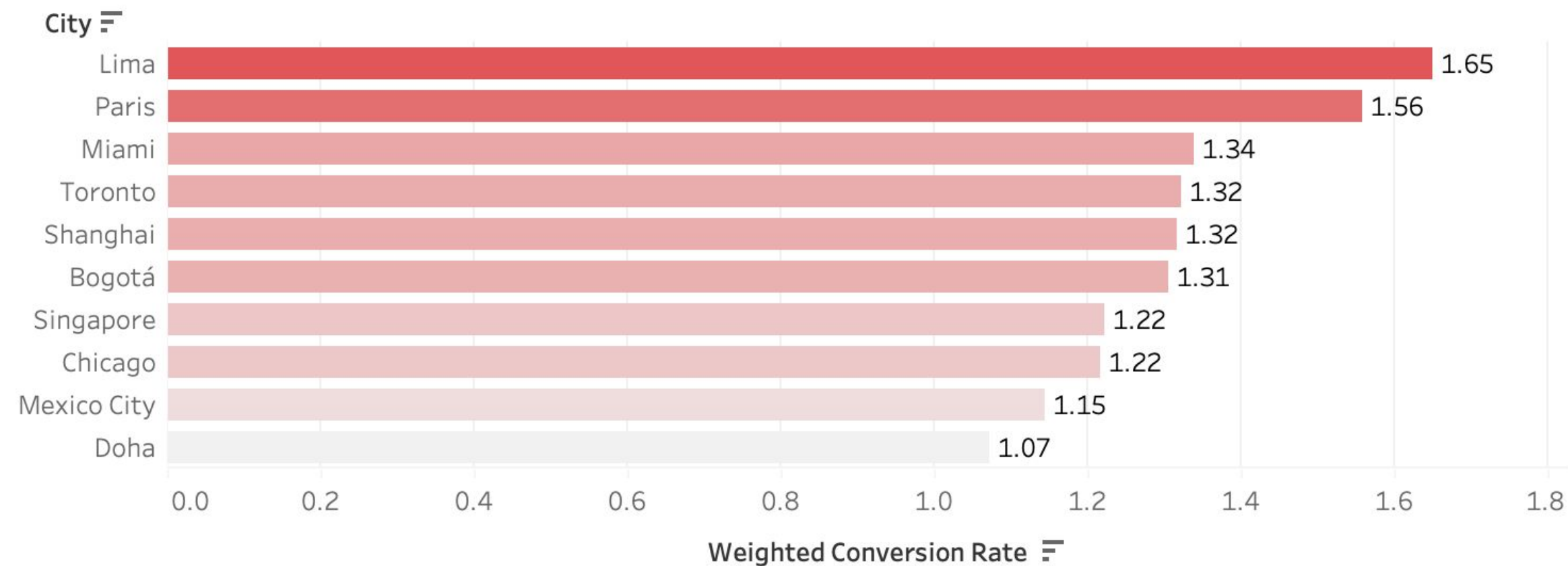
Conversion Analysis

Feb 2024 - Aug 2025

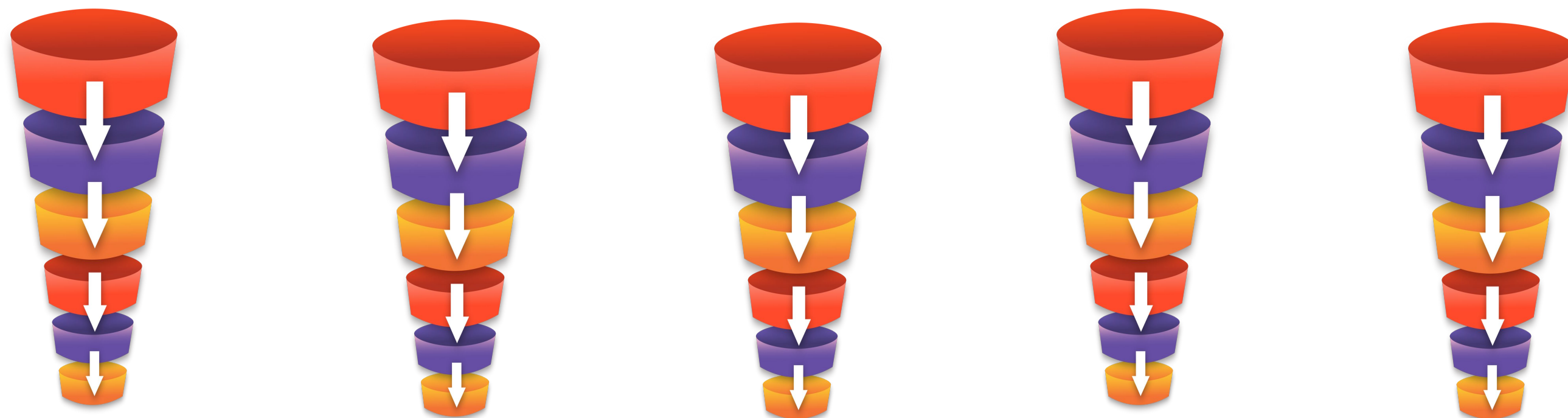
CONVERSION RATING ACROSS LOCATION TYPE



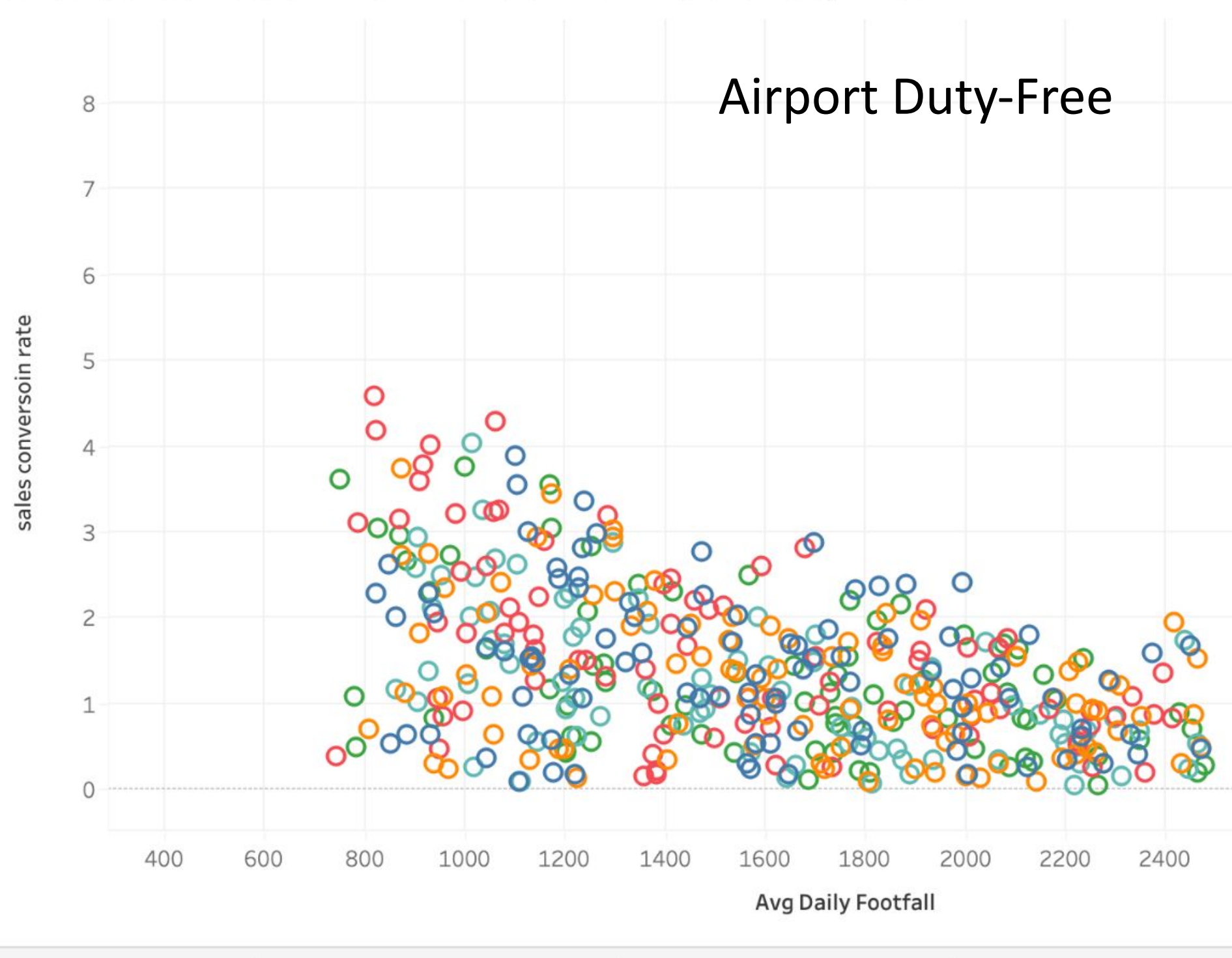
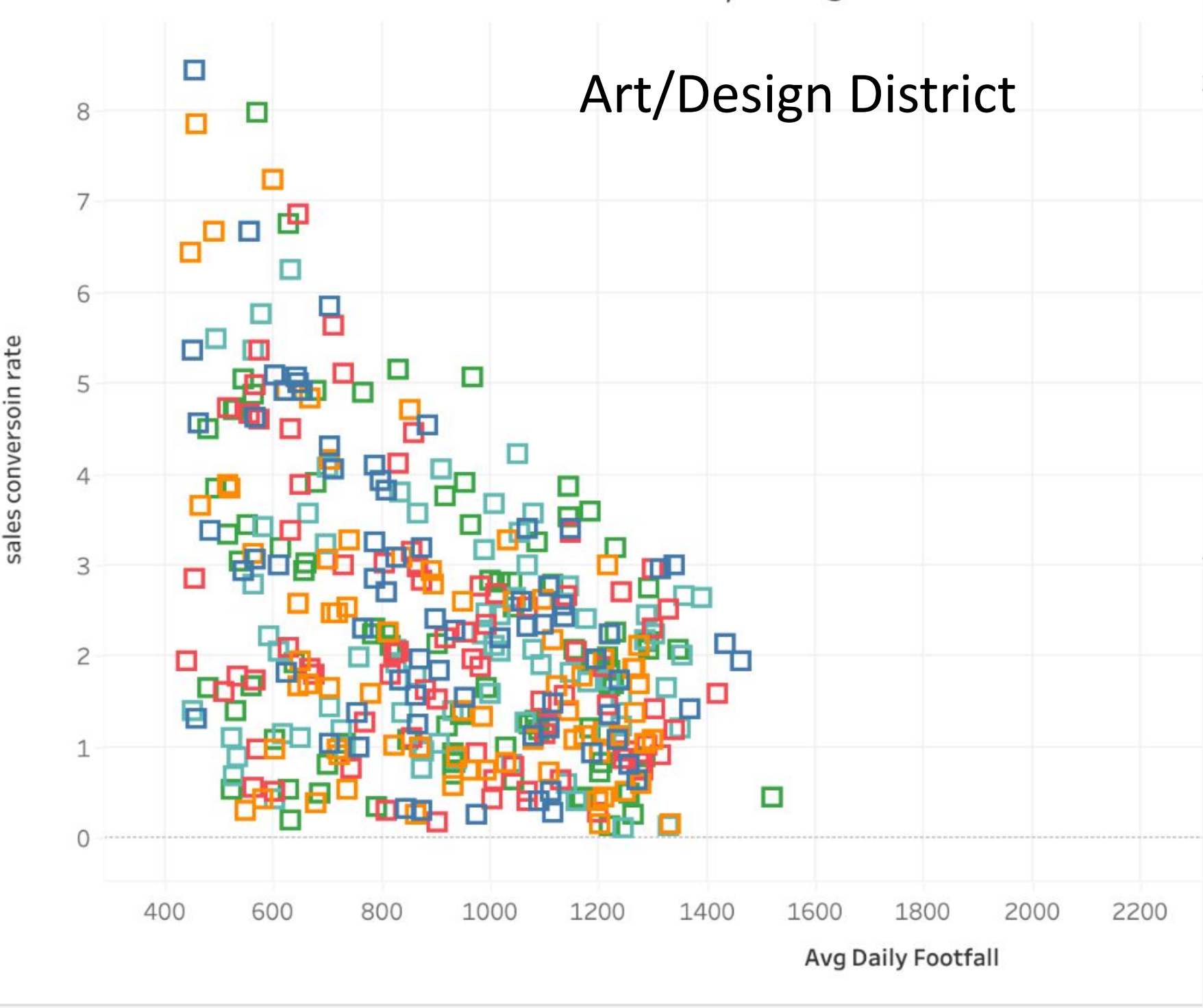
TOP AIRPORT CITIES



Footfall to Conversion



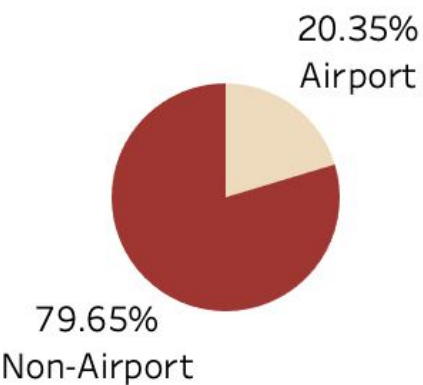
SALES CONVERSION RATE WITH TRAFFIC/AVG FOOTFALL



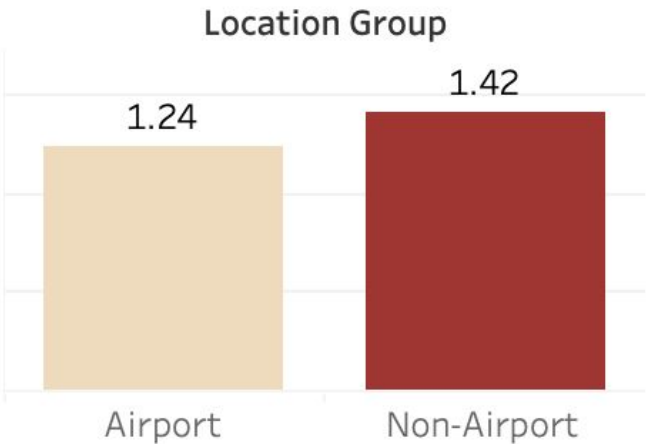
Airport Vs Non-airport Analysis

Total Revenue through Pop-up events = \$ 398.75 Million

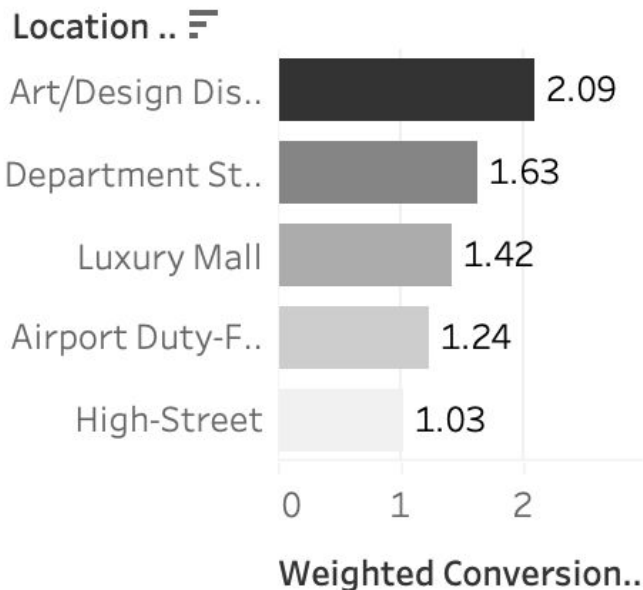
Event Distribution
2024-25



Weighted Conversion
Rate



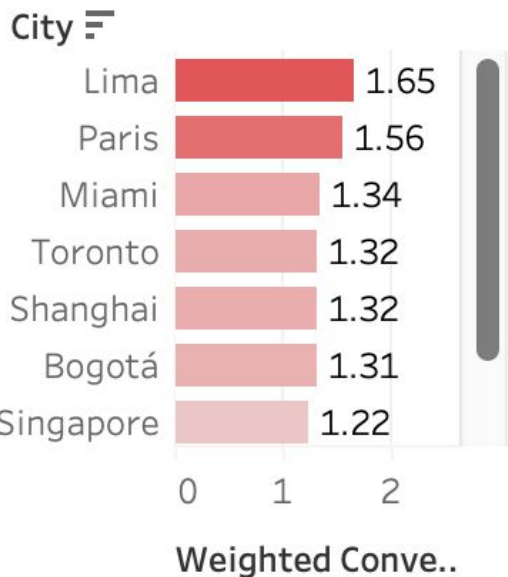
Location Type Ranking



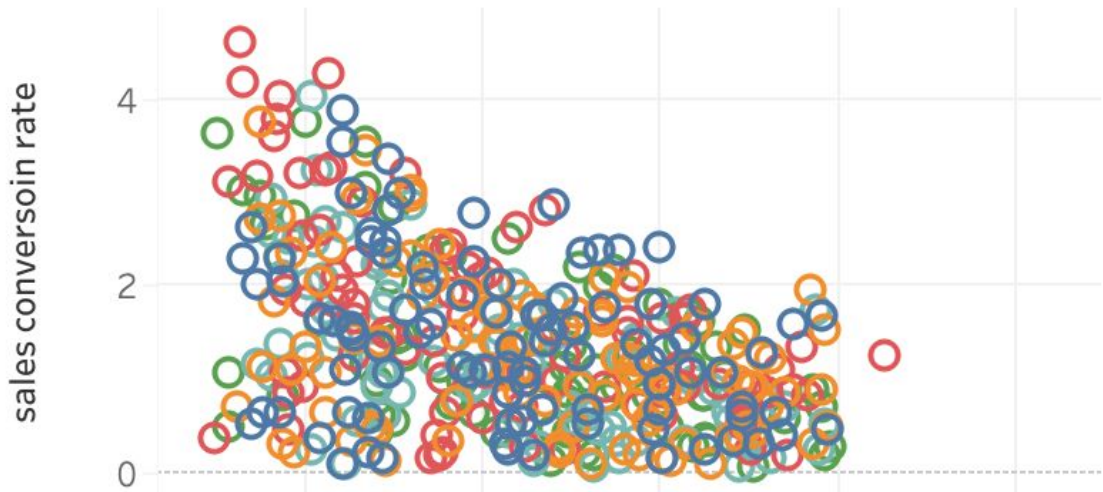
Location Type



Top Airport Cities



Sales Conversion Rate Vs Avg Footfall
(Location Type)



Avg Sell Through Percentage (Days-Lease wise)



DASHBOARD 1

Airport vs
Non-Airport
Conversion
Analysis

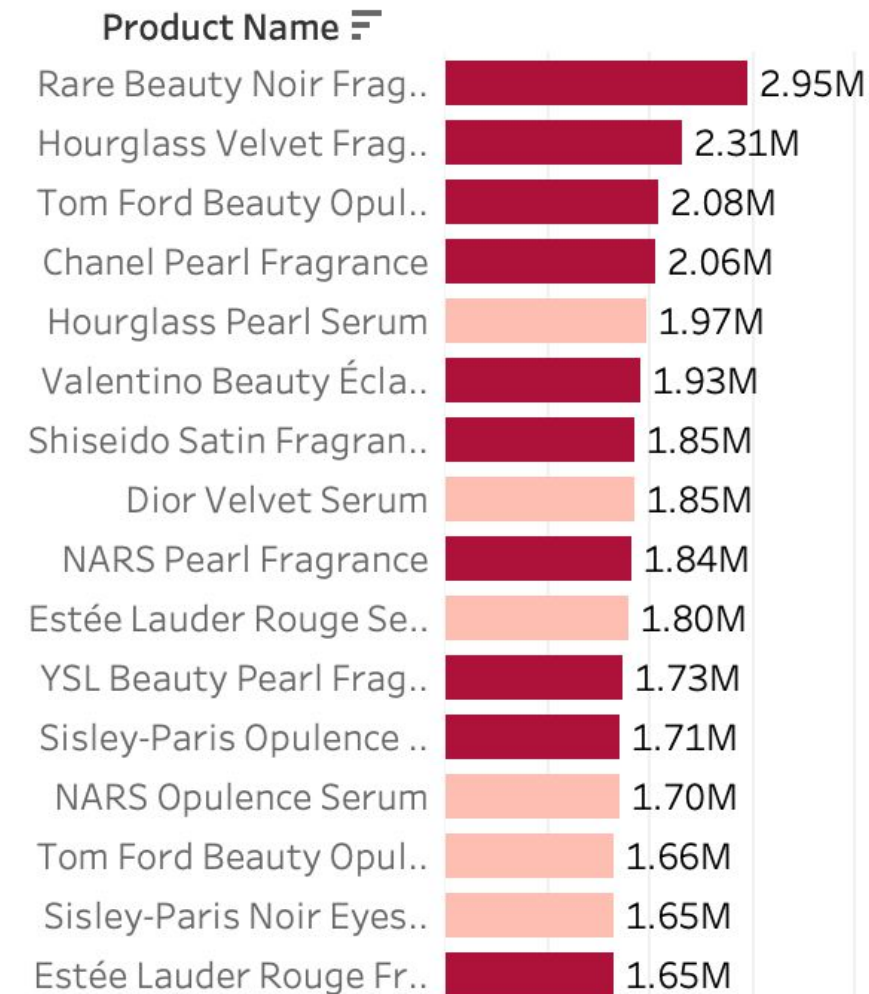
DASHBOARD 2

Product & Brand Performance Analysis

Feb 2024-Aug 2025

Product & Brand Performance Analysis 2024-2025

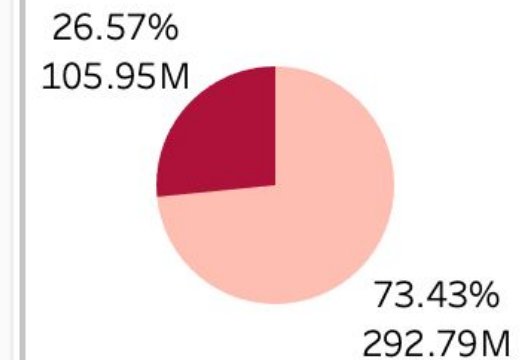
Top Products by Revenue



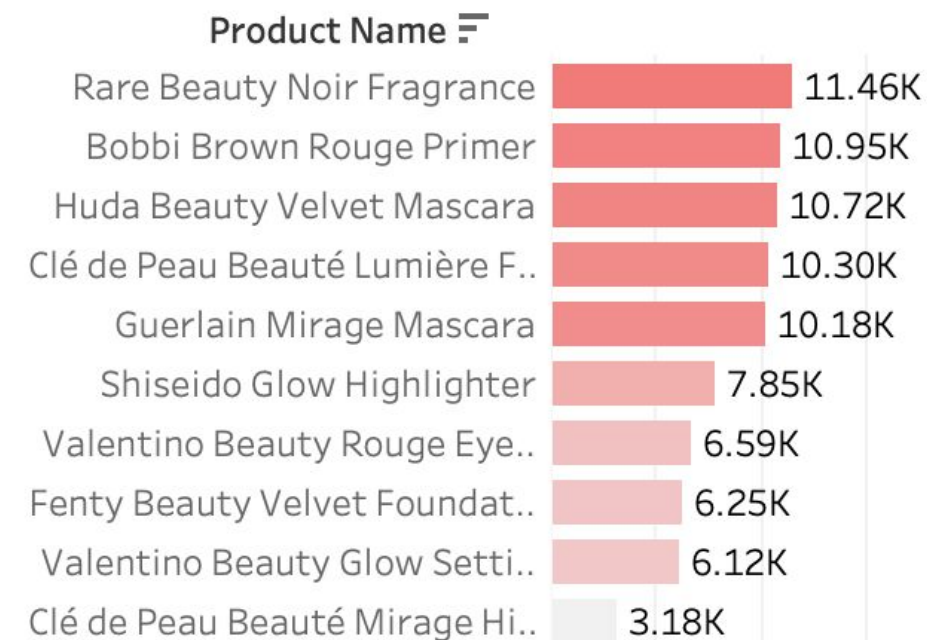
Product Type

- Cosmetics / Skin Care
- Fragrance

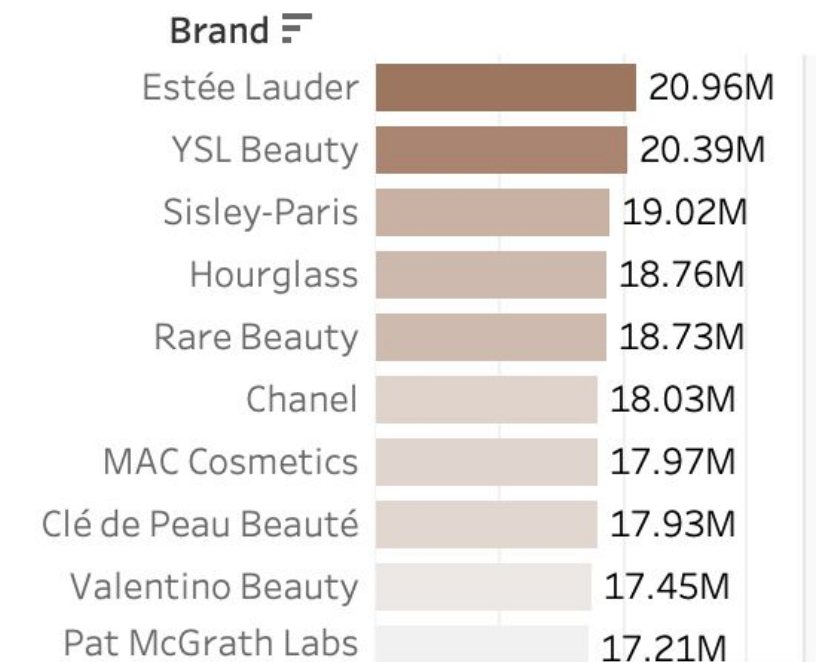
Revenue by Product Type



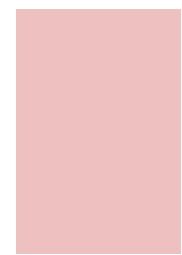
Top 10 Products by Units sold



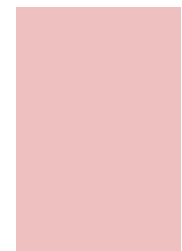
Top 10 Brands by Revenue



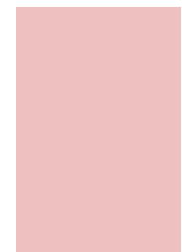
Key Insights



Non-airport outperforms airport duty-free location type in terms of conversion.



Higher Footfall is not equivalent to **higher conversion/revenue** especially in Airport duty free case.



Product-wise, **Fragrances** contribute to **majority of the revenue** (about 1/3rd of the total Revenue)



Airport Success is **city specific**, not universal



Art Districts and luxury malls with store atriums perform better



Almost every lease length(/duration of event) is contributing to the same range sell-through percentage value (60%-80%)



conclusion

- The ***hypothesis is accepted.***
- ***Non-airport luxury pop-up events are more effective*** at converting footfall into sales compared to Airport Duty-Free events, despite airports having higher traffic.

Opportunity Areas

**Prioritize luxury mall store
atriums and art districts
for majority of the events.**

**Focus more on the airport
primarily for brand
visibility, with
travel-exclusive products
in high-performing cities.**

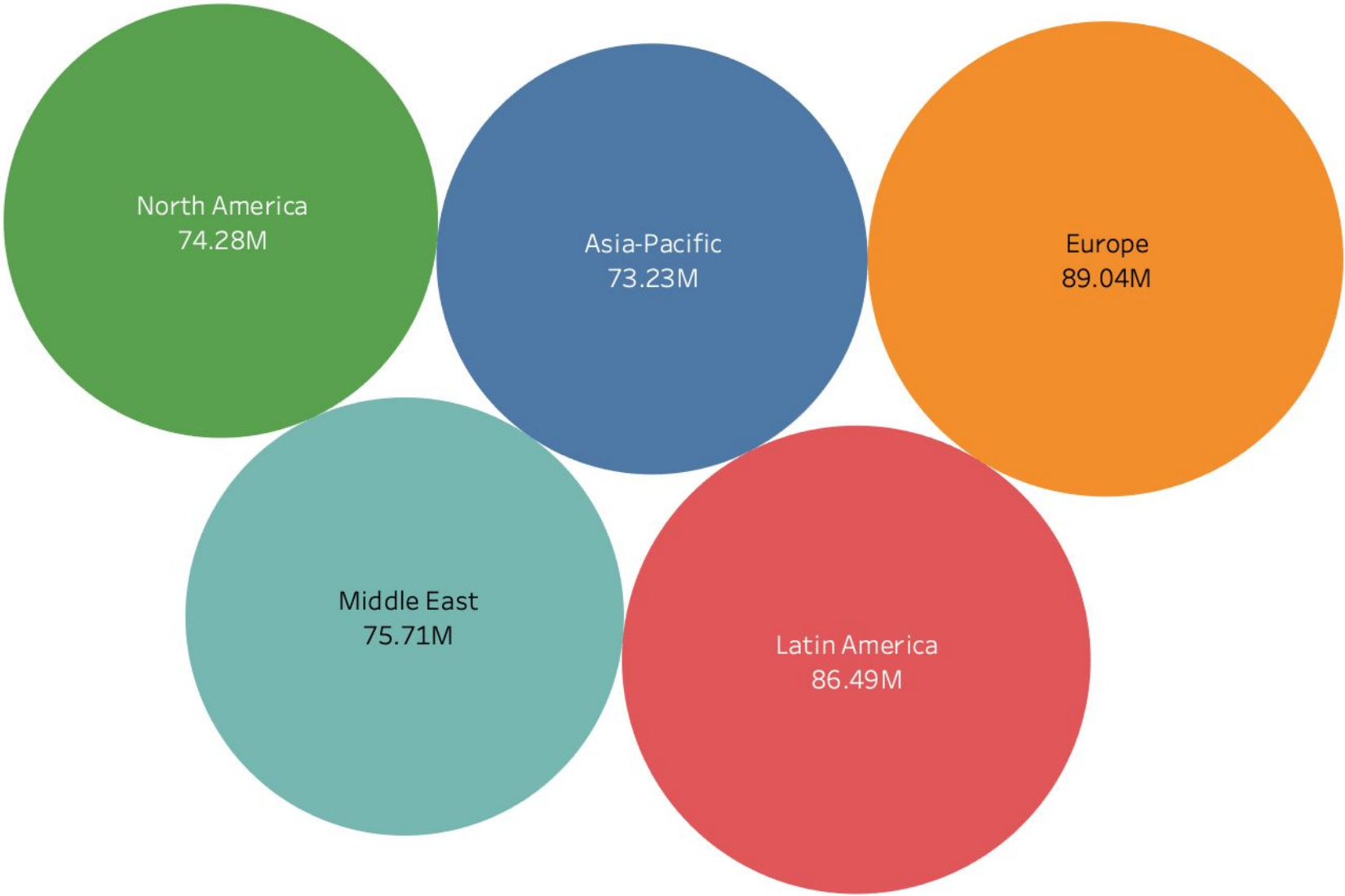
**Optimize the airport
product mix towards
fragrances.**

**Reconsider the time
duration for which the
events are run.**

Thank you

- Disclaimer: Dataset used for academic purposes only
 - Analysis conducted as part of coursework

Total Revenue Across Regions 2024-2025



No of Sales across Location Type 2024-2025

